Practice Assignment – IoT Fundamentals

MAIN TEST · Same format as *Individual assignment 1* · Coverage: Week 1–6 · Total: 100 marks

Name: _____

Student ID: _____

> **Practice version.** The structure, marks and question styles copy the teacher's assignment (Section A multiple choice, Section B short answer, Section C applied scenarios, Section D integrated communication), but **all questions are new**. Model answers and marking points are at the end.

Section A – Multiple choice (20 marks – 10 questions x 2 marks)

1. Which statement best describes the main idea of IoT?

A. Connecting as many devices as possible to Wi-Fi
B. Using data from physical things to know what is happening and to support decisions or actions
C. Replacing all computers with sensors
D. Storing every file in the cloud

2. A factory machine overheats and must trigger an alarm immediately. Where should the alarm decision be made?

A. On the device or at the edge
B. In the cloud, after one year of history is collected
C. By the maintenance engineer reading a monthly report
D. In a database backup

3. A sensor message contains only the value `29.7`. What is the main problem?

A. The value is too large to transmit
B. The value must be serialized to IPv6 first
C. Without a device ID, timestamp and unit, the value cannot be interpreted reliably
D. The value proves the sensor is broken

4. A program calculates a moving average with window 3 and another with window 10 on the same readings. Which statement is correct?

A. Window 10 reacts faster to sudden changes
B. Both windows give exactly the same result
C. Window 3 gives the smoother line
D. Window 3 reacts faster; window 10 gives a smoother result

5. A cold-chain box has a normal range of 2–8 °C and a physically valid range of −20 to 50 °C. A reading of 10 °C is:

A. INVALID
B. VALID, but TOO_WARM
C. VALID and NORMAL
D. INVALID and TOO_COLD

6. A radio transmits for 1 second every 10 seconds. What is its duty cycle?
   A. 1 %
   B. 0.1 %
   C. 10 %
   D. 90 %

7. A warehouse must identify many pallets at the same time without a direct line of sight. Which technology is most appropriate?
   A. RFID
   B. Barcode
   C. QR code
   D. Wi-Fi

8. Which statement about LoRa and LoRaWAN is correct?
   A. LoRa and LoRaWAN are exactly the same thing
   B. LoRaWAN is the radio modulation and LoRa is the network protocol
   C. LoRa is a cellular technology run by telecom operators
   D. LoRa is the radio modulation; LoRaWAN defines how devices, gateways and network services communicate

9. A UDP sender prints "UDP message sent", but the receiver program is not running. Which statement is correct?
   A. TCP will retransmit the datagram automatically
   B. UDP has no connection setup and no built-in delivery confirmation, so the sender does not know
   C. The receiver will get the datagram later when it starts
   D. The MQTT broker stores the datagram

10. Which topic is matched by the subscription `building/+/humidity`?
    A. `building/floor1/room2/humidity`
    B. `building/floor1/humidity`
    C. `building/humidity`
    D. `building/floor1/temperature`

Section B – Short Answer (30 marks – 5 questions x 6 marks)

11. Device, Edge and Cloud

A smart factory has vibration sensors on 200 machines. Explain the role of:

1. the device;
2. the edge; and
3. the cloud.

Give one suitable task for each in this factory.

12. Sensor data quality

A humidity sensor produces:

Time	Humidity
10:00	45.2 %
10:01	45.6 %
10:02	145.0 %
10:03	45.9 %
10:04	46.1 %

Answer:

a. What type of data is this?

b. Identify the data-quality problem.

c. Explain how the application could detect and handle the problem.

13. Duty Cycle and Battery

A soil sensor transmits for **2 seconds every 5 minutes**. While transmitting, the radio draws **40 mA**.

a. Calculate the duty cycle.

b. How many transmissions does it make per day, and what is the total transmit time per day?

c. Estimate the transmit energy per day in mAh ($E \approx I \times t$).

d. Give one design change that would extend battery life.

14. IPv6 and 6LoWPAN

Explain two differences between IPv4 and IPv6.

Then explain why a battery-powered IEEE 802.15.4 sensor still needs 6LoWPAN, even though IPv6 provides many addresses.

15. MQTT Quality of Service

Explain the difference between MQTT QoS 0, QoS 1 and QoS 2.

Your answer should refer to:

- the delivery guarantee;
- the messages exchanged;
- one suitable IoT message for each level.

Do not simply state that one level is "more reliable."

Section C – Applied Scenario Questions (40 marks – 2 questions x 20 marks)

16. Smart Parking System

A city installs battery-powered occupancy sensors in **3,000 street parking spaces** across the city centre.

Each sensor:

- sends a very small message (a few bytes) only when a space becomes free or occupied;
- is installed in the road surface, often far from any building;
- must operate for at least five years on one battery;
- has no Wi-Fi access point nearby.

The data is used by a driver parking app, a city traffic dashboard and a payment system.

a. Identify three communication requirements that are important when selecting connectivity for these sensors. (3 marks)

b. Choose a suitable connectivity technology.

Possible examples include:

- Wi-Fi
- BLE
- LoRa
- NB-IoT
- 4G/5G

Give two reasons for your choice. (5 marks)

c. A sensor sends:

```
{
  "space": "P-1043",
  "occupied": true,
  "battery": 87
}
```

If MQTT is used, propose a suitable **topic** for this message and explain the difference between the topic and the **payload**. (3 marks)

d. Three applications (app, dashboard, payment system) need the same occupancy data. Would request–response or publish–subscribe be more suitable?

Explain. (4 marks)

e. Which MQTT QoS level would you choose for the occupancy messages, and why?

The driver app is restarted and needs the **current** state of each space immediately, without waiting for the next change. Which MQTT feature helps, and how? (5 marks)

17. Hospital Ward Environment Monitor

A hospital places small battery-powered temperature and humidity sensors in every ward.

Each sensor:

- has very limited memory and processing capability;
- communicates over IEEE 802.15.4 in a **mesh** network;
- should participate in an IPv6-based IoT network;
- is occasionally asked for its latest reading by the nurse-station application.

A Python **edge gateway** in each ward also checks the temperature: if it is above **26 °C**, it publishes the command `AC ON` to the ward's air-conditioner controller.

a. Which application protocol is suitable for requesting a reading from these constrained sensors? (2 marks)

b. Which transport protocol is commonly used with this application protocol? (2 marks)

c. Give one advantage and one cost of using a **mesh** topology for these sensors. (3 marks)

d. Complete the protocol stack: (5 marks)

```
Ward Monitoring Application
        ↓
_____________________
        ↓
_____________________
        ↓
IPv6
        ↓
_____________________
        ↓
IEEE 802.15.4
```

e. Explain the role of the edge gateway. Your answer should refer to: (8 marks)

- why the edge gateway is both an MQTT subscriber and an MQTT publisher;
- which topic the air-conditioner controller should subscribe to;
- why the decision is made at the edge instead of in a remote cloud.

Section D — Integrated IoT Communication (10 marks)

18 — Trace the Data

Consider the following IoT system from the week 6 practical:

```
Humidity Sensor
↓
Python dictionary  {"device": "lab01", "humidity": 48.5}
↓
json.dumps()
↓
encode()
↓
TCP socket  (socket.AF_INET, socket.SOCK_STREAM)
↓
IP  (127.0.0.1, port 5000)
↓
TCP server  (recv → decode → json.loads)
```

Explain the role of each of the following:

Technology	Role
Python dictionary	
`json.dumps()`	
`encode()`	
`SOCK_STREAM` (TCP)	
`AF_INET` (IP)	
Port 5000	

What would change, and what would stay the same, if the program used `SOCK_DGRAM` instead?

Answer Key and Marking Guide

Section A (2 marks each)

Q	Ans	Why
1	B	IoT is about **using data from physical things**, not only connecting devices.
2	A	Urgent decisions stay local (device/edge) to avoid cloud latency and network dependency.
3	C	An IoT record needs device ID, timestamp, quantity, value and unit.
4	D	A small window responds faster; a large window filters short-term fluctuations.
5	B	10 °C is physically possible (VALID) but above 8 °C (TOO_WARM). Validation ≠ status.
6	C	1 ÷ 10 = 0.1 = 10 %.
7	A	RFID: radio, no line of sight, many tags at once.
8	D	LoRa = modulation (physical layer); LoRaWAN = network protocol.
9	B	"Message sent" does not prove the receiving application got it.
10	B	`+` matches **exactly one** level. A has two levels in the middle; C has none; D ends in `temperature`.

Section B (6 marks each)

11. (2 marks per part: role + task)

- **Device** – collects raw data close to the machine with very limited CPU/memory: reads vibration values, checks its battery, simple filtering.
- **Edge** – processes data **near the devices** (gateway/PLC in the factory): aggregates and filters readings from many machines, detects an abnormal vibration and raises an immediate alarm. Benefit: less bandwidth, faster response, keeps working if the Internet fails.
- **Cloud** – large-scale storage and analytics: stores months of history from all 200 machines, analyzes factory-wide trends, trains predictive-maintenance models, dashboards.

12.

a. **Time-series data** (sensor values recorded with timestamps at regular intervals). (2)

b. 145.0 % at 10:02 is **physically impossible** (humidity is 0–100 %): an **invalid value / outlier**, probably a sensor or transmission error. (2)

c. **Validate** each reading against the physically valid range (0–100 %) and mark it `INVALID`. Then do not use it in averages or alarms: discard it, keep it flagged, or replace it with `None` / leave a gap; a moving average over the valid values (≈ 45.6–45.9 %) also smooths noise. Do **not** silently replace it with 0. (2)

13.

a. Period = 5 min = 300 s. Duty cycle = 2 ÷ 300 = **0.67 %**. (1.5)

b. 24 h × 60 ÷ 5 = **288 transmissions/day**. Transmit time = 288 × 2 s = **576 s ≈ 0.16 h**. (1.5)

c. E ≈ 40 mA × 0.16 h = **6.4 mAh/day**. (1.5)

d. Any one: transmit less often (e.g. every 15 minutes, or only when the value changes); send less data (smaller/compact payload) so the radio is on for less time; filter or aggregate at the device; keep the radio asleep between transmissions. Lower duty cycle → longer battery life. (1.5)

14.

- IPv4 = **32-bit** address, dotted decimal (e.g. `192.168.1.25`), **limited** address space. IPv6 = **128-bit** address, hexadecimal in 8 colon-separated groups (e.g. `2001:db8::25`), an **extremely large** address space for very large numbers of IoT devices. (any two differences, 3 marks)

- IPv6 solves the **address shortage**, not the limits of constrained devices. A standard IPv6 header is **40 bytes**, too much overhead for tiny packets on a low-power 802.15.4 link with small frames, little memory and a battery. **6LoWPAN** is an **adaptation layer** that (1) **compresses the header to as few as 2–7 bytes**, (2) **fragments and reassembles** large IPv6 packets, and (3) supports **mesh routing**. A border router then connects the sensor network to the normal IPv6 Internet. (3 marks)

15. (2 marks per level)

QoS	Guarantee	Messages	Suitable message
0	**At most once**: no acknowledgment, no MQTT-level redelivery; a message can be lost	PUBLISH only	Periodic temperature telemetry: the next reading arrives soon anyway
1	**At least once**: publisher waits for an acknowledgment and resends if it is missing; **duplicates are possible**	PUBLISH → PUBACK	Fire alarm or machine-fault alarm: must arrive; receiving it twice is acceptable
2	**Exactly once**: four-step handshake prevents both loss and duplicates; highest overhead	PUBLISH → PUBREC → PUBREL → PUBCOMP	Critical control command or a payment/transaction message where duplicate processing must be avoided

The trade-off is reliability against speed and network cost.

Section C (20 marks each)

16. Smart Parking

a. Any three (1 each): **long range / coverage** (sensors spread across the city, far from buildings); **very low power** (5+ years on a battery); **very low data rate** (a few bytes, only on change); **good coverage through obstacles** (sensor in the road surface, cars parked on top); **many devices / density** (3,000 sensors); **no local infrastructure** (no Wi-Fi nearby).

b. (choice 1 mark + two reasons 2 marks each) **NB-IoT** is the best fit: it is a **cellular LPWAN** using the operator's existing network, so no gateways or Wi-Fi need to be installed across the city; it has **wide-area and good indoor/underground coverage**; **low power** for multi-year battery life; and **low data rate** is enough for a static device sending tiny messages (like smart meters). *Also accepted:* **LoRa** with a few private gateways: long range, very low power, very low data rate, private network without a subscription. Not suitable: Wi-Fi (no AP, high power), BLE (short range), 4G/5G (high power, far more bandwidth than needed).

c. Topic example: `city/parking/zoneA/P-1043/occupancy` (or `parking/<zone>/<space>/status`). (1) The **topic** is the label used by the broker to **classify and route** the message; the **payload** (the JSON) is the **actual information**. (2) A hierarchy lets the dashboard subscribe to `city/parking/#` or `city/parking/zoneA/+/occupancy`.

d. **Publish–subscribe** (1). With request–response every application would have to **poll** each of 3,000 sensors repeatedly, creating unnecessary traffic and poor scaling, and the sensor would need to know every destination. With pub-sub the sensor publishes **once to the broker**; the broker distributes to every subscriber; sensors and applications are **decoupled**, and a new application can be added just by subscribing. (3)

e. **QoS 1** (1): a status change happens rarely and must not be lost (drivers and payments depend on it), and receiving the same "occupied" twice does no harm; QoS 2's extra overhead is not needed (QoS 0 is risky because a lost change is not repeated soon). (2) **Retained message** (1): the sensor publishes its status with the retain flag; the broker keeps the **latest** message for each topic and sends it immediately to a new subscriber, so the restarted app gets the current state of every space at once (latest state only, not history). (1) *(Bonus: a **Last Will** of `offline` on a status topic tells the city when a sensor disconnects unexpectedly.)*

17. Hospital Ward Monitor

a. **CoAP** (Constrained Application Protocol): lightweight, request–response, resource-oriented, compact binary. (2)

b. **UDP**. (2)

c. Advantage (1.5): **extended coverage** (a message can hop through neighbours to reach the coordinator) and **alternative paths** if one node fails. Cost (1.5): **relay burden**: a node sends its own data and relays others', so more radio-on time and **more energy**; also more complexity.

d. (5) Ward Monitoring Application → **CoAP** → **UDP** → IPv6 → **6LoWPAN** → IEEE 802.15.4.

e. (8)

- The edge gateway **subscribes** to the sensor topic (e.g. `hospital/ward3/temperature`) to receive readings, parses the JSON payload and applies the rule *temperature > 26 °C →* `AC ON`*, otherwise* `AC OFF`. It then **publishes** the command to a **different topic**, so it is both subscriber and publisher. (3)
- The air-conditioner controller is a **subscriber to the command topic** (e.g. `hospital/ward3/ac/command`), **not** the sensor topic. The command is application-defined. (2)
- Deciding at the edge gives **low latency** (no round-trip to a distant cloud), keeps working if the **Internet connection fails**, saves **bandwidth** (only important events need to go to the cloud), and keeps patient-area data **local** for privacy. Complete loop: sense → communicate → process → act. (3)

Section D (10 marks)

Technology	Role
Python dictionary	The structured **application data** inside the program (key–value pairs: device, humidity). It cannot be sent as it is.
`json.dumps()`	**Serializes** the dictionary into a **JSON string**: a standard text representation that other programs can read.
`encode()`	Converts the text into **bytes**, because a socket transmits bytes, not strings.
`SOCK_STREAM` (TCP)	**Transport layer**: connection-oriented (the client must `connect()`; the server uses `listen()` / `accept()`), reliable, ordered **byte stream** delivered to the right process.
`AF_INET` (IP)	**Network layer**: IPv4 addressing; `127.0.0.1` identifies the **host** (localhost).
Port 5000	Identifies the **application/process** on that host (the TCP server).

(1.5 marks per row = 9)

With `SOCK_DGRAM` (UDP): no `connect()` / `accept()` and no handshake; the sender uses `sendto()` and the receiver `recvfrom()`; each send is a separate **datagram** (message boundaries preserved), with **no built-in delivery confirmation, ordering or retransmission**. The **JSON payload and its meaning stay exactly the same**; it still uses IP addresses and ports. Only the transport behavior changes. (1)